$\uparrow \downarrow \uparrow$ $\quad J>0$ interaction between nearest spins

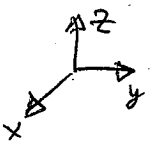
$\downarrow \uparrow \downarrow$ $\quad$ on a cubic lattice.

A sublattice: (up-spin)

$$\begin{cases} S_n^z = S - a_n^+ a_n \\ S_n^+ \cong \sqrt{2S}\, a_n \\ S_n^- \cong \sqrt{2S}\, a_n^+ \end{cases}$$

B sublattice: (down spin)

can obtain by rotation about $\hat{x}$ axis




$$\left.\begin{array}{l} S^x \to S^x \\ S^y \to -S^y \\ S^z \to -S^z \end{array}\right\} \quad \begin{array}{l} S^+ = S^x + iS^y \to S^x - iS^y = S^- \\ S^- = S^x - iS^y \to S^x + iS^y = S^+ \end{array} \Rightarrow \begin{cases} S_m^z = -S + b_m^+ b_m \\ S_m^+ = \sqrt{2S}\, b_m^+ \\ S_m^- = \sqrt{2S}\, b_m \end{cases}$$

$$H = \frac{1}{2} J \sum_{nm} \vec{S}_n \cdot \vec{S}_m = J \sum_{\substack{n \in A \\ (n+g = m \in B)}} \vec{S}_n \cdot \vec{S}_{n+g} =$$

$$= J \sum_{n,g} S_n^z S_{n+g}^z + \frac{1}{2}\left(S_n^+ S_{n+g}^- + S_n^- S_{n+g}^+\right) =$$

$$= J \sum_{n,g} \left(S - a_n^+ a_n\right)\left(-S + b_{n+g}^+ b_{n+g}\right) + \frac{1}{2}(2S)\left(a_n b_{n+g} + a_n^+ b_{n+g}^+\right) = J \sum_{ng} \Big\{ -S^2 + S\left(a_n^+ a_n + b_{n+g}^+ b_{n+g} + a_n b_{n+g} + a_n^+ b_{n+g}^+\right) - \underbrace{a_n^+ a_n b_{n+g}^+ b_{n+g}}_{\text{neglect for now}} \Big\}$$

$$\sum_{ng} (-S^2) = (-S^2)\left(\frac{N}{2}\right)(z)$$

$\frac{N}{2}$: # sites in A sublattice

$z$: # neighbors $\begin{cases} 6 & \text{for } d=3 \quad \text{cubic} \\ 4 & \text{for } d=2 \quad \text{square} \\ 2 & \text{for } d=1 \quad \text{chain} \end{cases} = \underline{2d}$

Fourier transform: $a_n = \sqrt{\frac{2}{N}} \sum_k e^{-i\vec{k}\vec{R}_n} a_k$

$$b_m = \sqrt{\frac{2}{N}} \sum_k e^{-i\vec{k}\vec{R}_m} b_k$$

where $\sum_k$ is over the magnetic Brillouin zone = Brillouin zone of a lattice formed by A-sublattice only. (consists of $N/2$ sites)

$$H^{(2)} = JS \sum_{n g} \frac{2}{N} \sum_{k_1 k_2} \Big\{ a^+_{k_1} a_{k_2} e^{i\vec{k}_1\vec{R}_n - i\vec{k}_2\vec{R}_n} + b^+_{k_1} b_{k_2} e^{i\vec{k}_1(\vec{R}+\vec{g}) - i\vec{k}_2(\vec{R}+\vec{g})}$$

$$+ a^+_{k_1} b^+_{k_2} e^{i\vec{k}_1\vec{R}_n} e^{i\vec{k}_2(\vec{R}_n+\vec{g})} + a_{k_1} b_{k_2} e^{-i\vec{R}_n(\vec{k}_1+\vec{k}_2)} e^{-i\vec{k}_2\vec{g}} \Big\}$$

$$= JS \sum_g \sum_k \Big\{ a^+_k a_k + b^+_k b_k + a^+_k b^+_{-k} e^{-i\vec{k}\vec{g}} + a_k b_{-k} e^{i\vec{k}\vec{g}} \Big\}$$

$$\sum_{R_n} e^{i\vec{R}_n(\vec{k}_1 \pm \vec{k}_2)} = \frac{N}{2} \begin{cases} \Delta_{k_1+k_2,0} \\ \Delta_{k_1-k_2,0} \end{cases}.$$

anomalous terms (involve two creation/annihilation operators at once)

$$H^{(2)} = JS \sum_k A_k (a^+_k a_k + b^+_k b_k) + B_k (a^+_k b^+_{-k} + a_{-k} b_k)$$

Here $A_k = z = 2d$ ; $B_k = \sum_g e^{-i\vec{k}\vec{g}} = 2(\cos k_x a + \cos k_y a + \cos k_z a) = B_{-k}$

<u>Bogoliubov transformation</u>

$$a_k = ch\theta_k \, \alpha_k - sh\theta_k \, \beta^+_{-k} \qquad \text{where } [\alpha_k, \alpha^+_{k'}] = \delta_{kk'}$$

$$b_k = -sh\theta_k \, \alpha^+_k + ch\theta_k \, \beta_{-k} \qquad [\beta_k, \beta^+_{k'}] = \delta_{k,k'}$$

$$[\alpha_k, \beta_{k'}] = 0.$$

Check $[a_k, a^+_{k'}] = [ch\theta_k \, \alpha_k - sh\theta_k \, \beta^+_{-k}, ch\theta_{k'} \, \alpha^+_{k'} - sh\theta_{k'} \, \beta^-_{-k'}]$

$$= ch^2\theta_k \underbrace{[\alpha_k, \alpha^+_{k'}]}_{\delta_{k,k'}} + sh^2\theta_k \underbrace{[\beta^+_{-k}, \beta_{-k'}]}_{-\delta_{k,k'}} = \underbrace{(ch^2\theta_k - sh^2\theta_k)}_{1} \delta_{k,k'}$$

$$\frac{H^{(2)}}{JS} = \sum_k A_k \Big[ (ch\theta_k \alpha_k^+ - sh\theta_k \beta_{-k})(ch\theta_k \alpha_k - sh\theta \beta_{-k}^+)$$

$$+ (-sh\theta_k \alpha_k + ch\theta_k \beta_{-k}^+)(-sh\theta_k \alpha_k^+ + ch\theta_k \beta_{-k}) \Big]$$

$$+ B_k \Big[ (ch\theta_k \alpha_k^+ - sh\theta_k \beta_{-k})(-sh\theta_k \alpha_k + ch\theta_k \beta_{-k}^+) +$$

$$+ (ch\theta_k \alpha_k - sh\theta_k \beta_{-k}^+)(-sh\theta_k \alpha_k^+ + ch\theta_k \beta_{-k}) \Big] =$$

$$= \sum_k \Big[ A_k (ch^2\theta_k + sh^2\theta_k) - B_k ch\theta_k sh\theta_k (\alpha_k^+\alpha_k + \alpha_k\alpha_k^+) \Big]$$

$$+ \Big[ A_k (sh^2\theta_k + ch^2\theta_k) - B_k sh\theta_k ch\theta_k (\beta_{-k}\beta_{-k}^+ + \beta_{-k}^+\beta_{-k}) \Big] +$$

$$- \alpha_k^+ \beta_{-k}^+ \Big[ 2A_k ch\theta_k sh\theta_k - B_k (ch^2\theta_k + sh^2\theta_k) \Big]$$

$$- \alpha_k \beta_{-k} \Big[ 2A_k ch\theta_k sh\theta_k - B_k (sh^2\theta_k + ch^2\theta_k) \Big] .$$

We <u>require</u> the anomalous terms $\alpha_k^+ \beta_{-k}^+$, $\alpha_k \beta_{-k}$ to <u>be absent</u>:

$$\Rightarrow \quad A_k sh 2\theta_k - B_k ch 2\theta_k = 0$$

$$\Rightarrow \quad \frac{sh 2\theta_k}{ch 2\theta_k} = \frac{B_k}{A_k} .$$

$$\Rightarrow H^{(2)}/JS = \sum_k \Big( A_k ch 2\theta_k - \tfrac{1}{2} B_k sh 2\theta_k \cdot 2 \Big) \alpha_k^+ \alpha_k - \tfrac{1}{2} B_k sh 2\theta_k$$

$$+ \Big( A_k ch 2\theta_k - B_k sh 2\theta_k \Big) \beta_{-k}^+ \beta_{-k} - \tfrac{1}{2} B_k sh 2\theta_k$$

$$- 2A_k sh^2\theta_k$$

$\alpha_k \alpha_k^+ = 1 + \alpha_k^+ \alpha_k$

Let $\omega_k = A_k ch 2\theta_k - B_k sh 2\theta_k$

$$ch^2 2\theta - sh^2 2\theta = 1 \Rightarrow \frac{sh\, 2\theta_k}{\sqrt{1+sh^2 2\theta_k}} = \frac{B_k}{A_k} \quad \left(= \text{even function of } \vec{k}\right)$$

$$\frac{sh^2 2\theta_k}{1+sh^2 2\theta_k} = \frac{B_k^2}{A_k^2} \Rightarrow \left(1-\frac{B_k^2}{A_k^2}\right) sh^2 2\theta_k = \frac{B_k^2}{A_k^2}$$

$$\Rightarrow sh^2 2\theta_k = \frac{B_k^2}{A_k^2 - B_k^2}$$

$$\Rightarrow ch^2 2\theta_k = 1 + \frac{B_k^2}{A_k^2 - B_k^2} = \frac{A_k^2}{A_k^2 - B_k^2}$$

$$sh\, 2\theta_k = sgn(B_k) \frac{|B_k|}{\sqrt{A_k^2 - B_k^2}}, \quad B_k\, sh\, 2\theta_k = \frac{B_k^2}{\sqrt{A_k^2 - B_k^2}}$$

$$ch\, 2\theta_k = \frac{A_k}{\sqrt{A_k^2 - B_k^2}}$$

$$\omega_k = \frac{A_k^2 - B_k^2}{\sqrt{A_k^2 - B_k^2}} = \sqrt{A_k^2 - B_k^2}$$

Constant term: $2A_k\, sh^2\theta_k - B_k\, sh\, 2\theta_k = A_k\, ch\, 2\theta_k - B_k\, sh\, 2\theta_k - A_k$

$$= \omega_k - A_k$$

$$2\, sh^2\theta_k = (ch\, 2\theta_k - 1)$$

$$\Rightarrow H^{(2)} = JS \sum_k \left\{ \omega_k \left(\alpha_k^+ \alpha_k + \beta_k^+ \beta_k\right) + \omega_k - A_k \right\}$$

$$= JS \sum_k \left\{ \omega_k \left(\alpha_k^+ \alpha_k + \tfrac{1}{2}\right) + \omega_k \left(\beta_k^+ \beta_k + \tfrac{1}{2}\right) \right\}.$$

energy of zero-point motion

$$\underbrace{- JS \sum_k A_k}$$

correction to the ground state energy $H^{(0)}$

$$H^{(0)} + H^{(2)} = -J\, S(S+1) \frac{N}{2} \cdot z + JS \sum_k \omega_k \left(\alpha_k^+ \alpha_k + \tfrac{1}{2}\right) + \omega_k \left(\beta_k^+ \beta_k + \tfrac{1}{2}\right)$$

<u>Two</u> branches of spin waves : $\alpha_k$ and $\beta_k$.

$$\omega_k^2 = A_k^2 - B_k^2 = (2d)^2 - \left(2\sum_{i=1}^{d} \cos k_i a\right)^2 =$$

$$= 4d^2 \left\{ 1 - \left( \frac{1}{d} \sum_{i=1}^{d} \cos k_i a \right)^2 \right\}$$

<u>$d=3$</u> $\quad \omega_k = 6\sqrt{1 - \left(\frac{1}{3}(\cos k_x a + \cos k_y a + \cos k_z a)\right)^2}$

Minimum near $\vec{k} = \vec{0}$

$\cos k_i a \cong 1 - \frac{1}{2} k_i^2 a^2$

$$\omega_k \cong 6\sqrt{1 - \frac{1}{9}\left(3 - \frac{a^2}{2}(k_x^2 + k_y^2 + k_z^2)\right)^2} =$$

$$= 6\sqrt{1 - \left(1 - \frac{a^2}{6}\vec{k}^2\right)^2} = 6\sqrt{1 - 1 + \frac{2a^2}{6}\vec{k}^2} = \sqrt{12 a^2 \vec{k}^2}$$

$$= 12a\,|\vec{k}|.$$

! * Spin-wave dispersion is linear in momentum ("relativistic"!)

Low-energy excitations: $\sum_k 12a\,JS\,|\vec{k}|\,(\alpha_k^\dagger \alpha_k + \beta_k^\dagger \beta_k)$.

Ground state energy is obtained by taking expectation value $E_g = \langle H^{(0)} + H^{(2)} \rangle = -JS^2 \frac{N}{2} z - JS \frac{N}{2} z$

$$+ JS \sum_k \omega_k$$

because $\langle 0 | \alpha_k^\dagger \alpha_k | 0 \rangle = 0$ as $\alpha_k |0\rangle = 0$, $\beta_k |0\rangle = 0$.

$$E_g = -JS^2 \frac{N}{2} z \left\{ 1 + \frac{1}{S} - \frac{2}{SNz} \sum_k \omega_k \right\} =$$

$$= -JS^2 \frac{Nz}{2} \left\{ \underset{\substack{\uparrow \\ \text{classical}}}{1} + \frac{1}{S} \underbrace{\left[ 1 - \frac{2}{N} \sum_k \frac{\omega_k}{z} \right]}_{\text{quantum (first } \frac{1}{S} \text{ correction)}} \right\}$$

$$E_g = -\frac{1}{2} J S^2 N z \left\{ 1 + \frac{1}{S}\left[1 - \frac{2}{N}\sum_k \sqrt{1-\left(\frac{1}{d}\sum \cos k_i a\right)^2}\right]\right\}$$

Spin expectation value $\langle S_n \rangle = S - \langle a_n^+ a_n \rangle =$

$$= S - \frac{2}{N}\sum_{k_1 k_2} e^{i(\bar{k}_1 - \bar{k}_2)\bar{R}_n} \langle a_{k_1}^+ a_{k_2} \rangle$$

$$\langle a_{k_1}^+ a_{k_2} \rangle = \langle (\mathrm{ch}\,\theta_{k_1} \alpha_{k_1}^+ - \mathrm{sh}\,\theta_{k_1} \beta_{-k_1})(\mathrm{ch}\,\theta_{k_2} \alpha_{k_2} - \mathrm{sh}\,\theta_{k_2} \beta_{-k_2}^+) \rangle$$

$$= \mathrm{ch}\,\theta_{k_1} \mathrm{ch}\,\theta_{k_2} \langle \alpha_{k_1}^+ \alpha_{k_2} \rangle + \mathrm{sh}\,\theta_{k_1} \mathrm{sh}\,\theta_{k_2} \langle \beta_{-k_1} \beta_{-k_2}^+ \rangle$$

$$- \mathrm{ch}\,\theta_{k_1} \mathrm{sh}\,\theta_{k_2} \underbrace{\langle \alpha_{k_1}^+ \beta_{-k_2}^+ \rangle}_{=0} - \mathrm{sh}\,\theta_{k_1} \mathrm{ch}\,\theta_{k_2} \underbrace{\langle \beta_{-k_1} \alpha_{k_2} \rangle}_{=0}$$

no anomalous terms in the ground state of $\alpha, \beta$.

$\langle \alpha_{k_1}^+ \alpha_{k_2} \rangle = 0$ because $\alpha_{k_2}|0\rangle = 0$ for any $k_2$.

But $\langle \beta_{-k_1} \beta_{-k_2}^+ \rangle = \delta_{k_1,k_2} \langle \underbrace{\beta_{-k_1} \beta_{-k_1}^+}_{1+\beta_{-k_1}^+ \beta_{-k_1}} \rangle = \delta_{k_1,k_2}$

$$\Rightarrow \langle S_n \rangle = S - \frac{2}{N}\sum_k \mathrm{sh}^2\theta_k = S - \frac{2}{N}\sum_k \frac{1}{2}(\mathrm{ch}\,2\theta_k - 1)$$

$$= S - \frac{2}{N}\sum_k \frac{1}{2}\left(\frac{A_k}{\omega_k} - 1\right) = S\left\{ \underbrace{1}_{\text{classical result}} - \frac{1}{2S} \cdot \underbrace{\frac{2}{N}\sum_k \left(\frac{1}{\sqrt{1-\gamma_k^2}} - 1\right)}_{\text{quantum correction } (1/S)} \right\}$$

$$\gamma_k \equiv \frac{1}{d}\sum_{i=1}^{d} \cos k_i a$$

Thus in $d=2$, for example, $\langle S_n \rangle = S\left\{1 - \frac{1}{2S}\cdot(1.393-1)\right\} =$

$= S\left(1 - \frac{0.393}{2S}\right)$ so that for $S = \frac{1}{2}$ the ~~[illegible]~~ reduction is

$\langle S_n \rangle = \frac{1}{2}(0.607) = 0.303$

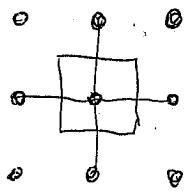
$\boxed{\text{about 40\% reduction}}$ instead of classical value $\frac{1}{2} = 0.5$

via Mathematica

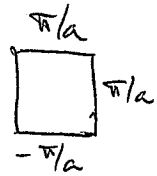
$$I_1 = \frac{2}{N}\sum_k \sqrt{1-\cos^2 k} = \int_{-\pi}^{\pi} \frac{dk}{2\pi}\,|\sin k| = \frac{2}{2\pi}\int_0^{\pi} dk \sin k =$$

$$= \frac{-1}{\pi}\cos k\Big|_0^{\pi} = \frac{2}{\pi}.$$

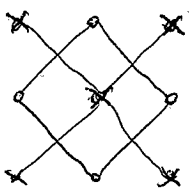
$$I_2 = \frac{2}{N}\sum_k \sqrt{1-\left(\frac{1}{2}(\cos k_x + \cos k_y)\right)^2} = \frac{2a^2}{(2\pi)^2}\int_{MBZ} dk_x\, dk_y \sqrt{1-\frac{1}{4}(\cos k_x + \cos k_y)^2}$$



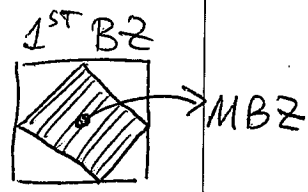
$\Rightarrow$




$\Rightarrow \left(\frac{2\pi}{a}\right)^2$



$$\left(\frac{2\pi}{\sqrt{2}a}\right)^2 = \frac{1}{2}\left(\frac{2\pi}{a}\right)^2$$




can work with full BZ instead
$\Downarrow$
simpler boundaries for numerical evaluation

$$I_2 = \frac{1}{(2\pi)^2}\int_{-\pi}^{\pi} dx \int_{-\pi}^{\pi} dy \sqrt{1-\frac{1}{4}(\cos x + \cos y)^2} = \frac{4}{(2\pi)^2}\int_0^{\pi} dx \int_0^{\pi} dy$$

$$\sqrt{1-\frac{1}{4}(\cos x + \cos y)^2} = (1 - 0.158) = 0.842$$

$$I_3 = \frac{1}{(2\pi)^3}\int dx\,dy\,dz \sqrt{1-\frac{1}{9}(\cos x + \cos y + \cos z)^2} = \frac{1}{\pi^3}\int_0^{\pi} dx \int_0^{\pi} dy \int_0^{\pi} dz \sqrt{\phantom{x}}$$

$$= 0.903 = 1 - 0.097.$$

Then

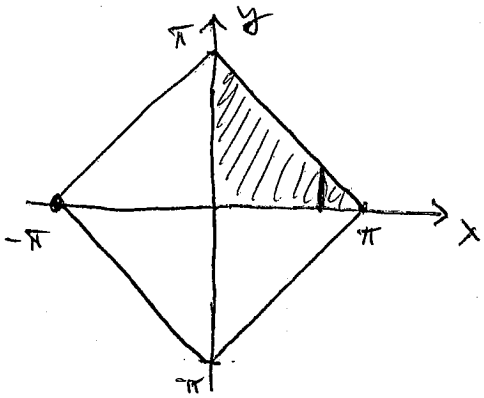
$$J_2 = \frac{1}{(2\pi)^2}\int_{-\pi}^{\pi} dx\,dy \frac{1}{\sqrt{1-\frac{1}{4}(\cos x + \cos y)^2}} = \frac{4}{(2\pi)^2}\int_0^{\pi} dx\,dy \frac{1}{\sqrt{\phantom{x}}} = 1.393$$

$$J_3 = \frac{1}{(2\pi)^3}\int_{-\pi}^{\pi} dx\,dy\,dz \frac{1}{\sqrt{1-\frac{1}{9}(\cos x + \cos y + \cos z)^2}} = 1.156$$

$$J_1 = \frac{2}{\pi}\ln\left(\tan\frac{x}{2}\right)\Big|_0^{\pi/2}$$

$$J_1 = \int_{-\pi}^{\pi} \frac{dx}{2\pi}\frac{1}{\sqrt{1-\cos^2 x}} = \frac{1}{\pi}\int_0^{\pi} \frac{dx}{\sin x} = \infty!$$ diverges at $x = 0, \pi$. logarithmically.

$$I_2 = \frac{2}{(2\pi)^2} \int_{MBZ} dx\,dy \sqrt{1 - \frac{1}{4}(\cos x + \cos y)^2}$$




$\left.\begin{array}{l} x \to -x \\ y \to -y \end{array}\right\}$ integrate over $\frac{1}{4}$ of MBZ

$$I_2 = \frac{2 \cdot 4}{(2\pi)^2} \int_0^{\pi} dx \int_0^{\pi - x} dy \sqrt{1 - \frac{1}{4}(\cos x + \cos y)^2} = 2 \cdot 0.421 = 0.842$$

as before.

$$\frac{2}{\pi^2} \int_0^{\pi} dx\,(\pi - x) = \frac{2}{\pi^2}\left(\pi^2 - \frac{1}{2}x^2\Big|_0^{\pi}\right) = \frac{2}{\pi^2} \cdot \frac{\pi^2}{2} = 1$$

Problems: (1) Thermodynamics of ordered AFM state

(2) $J_1 - J_2$ model : phase diagram as function of $J_2/J_1$ ratio.